

YouTube Analytics

SQL Project — Query Results & Data Report
June 2025

20 Total Users	20 Channels	4.0M Total Views	15 Comments
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Database	Relational SQL (7 tables)
Date Range	January 2023 – April 2025
Countries	USA, India, UK, Canada, Australia, Brazil, France, Germany, Italy, UAE, Japan
Categories	Technology, Education, Gaming, Travel, Music, Fitness, Editing, Entertainment, Food, Creativity, Lifestyle, Fashion, Reviews, Cooking, Streaming
Top Video	Top 10 Games — 1,200,000 views
Top Channel	Leo Gamer Pro — 245,000 subscribers

1. Database Tables — Raw Data

1.1 Users (20 rows)

Registered users on the platform from 11 countries, joined between Jan 2023 and Apr 2024.

user_id	username	country	join_date
1	john_doe	USA	2023-01-10
2	priya_tech	India	2023-02-15
3	alex_gaming	UK	2023-03-20
4	sarah_vlogs	Canada	2023-04-18
5	mike_music	Australia	2023-05-25
6	emma_fit	USA	2023-06-12
7	raj_edits	India	2023-07-08
8	lucas_live	Brazil	2023-08-21
9	nina_foodie	France	2023-09-14
10	david_creator	Germany	2023-10-01

user_id	username	country	join_date
11	oliver_tech	UK	2023-10-19
12	kavya_vlogs	India	2023-11-03
13	sam_music	USA	2023-11-25
14	leo_gamer	Canada	2023-12-10
15	mia_travel	Australia	2024-01-05
16	arjun_sql	India	2024-01-22
17	sofia_style	Italy	2024-02-11
18	ethan_reviews	USA	2024-02-28
19	zara_cooking	UAE	2024-03-15
20	noah_streams	Japan	2024-04-02

1.2 Channels (20 rows)

Each user owns exactly one channel. Subscriber counts range from 54,000 to 245,000.

channel_id	user_id	channel_name	category	subscribers
101	1	John Tech Reviews	Technology	120000
102	2	Priya Coding Hub	Education	95000
103	3	Alex Gaming Zone	Gaming	210000
104	4	Sarah Travel Vlogs	Travel	150000
105	5	Mike Music Studio	Music	175000
106	6	Emma Fitness World	Fitness	88000
107	7	Raj Editing Master	Editing	67000
108	8	Lucas Live Streams	Entertainment	143000
109	9	Nina Food Recipes	Food	99000
110	10	David Creative Studio	Creativity	76000
111	11	Oliver Tech Talks	Technology	134000
112	12	Kavya Daily Vlogs	Lifestyle	112000
113	13	Sam Music Beats	Music	156000
114	14	Leo Gamer Pro	Gaming	245000
115	15	Mia Travel Diaries	Travel	98000

channel_id	user_id	channel_name	category	subscribers
116	16	Arjun SQL Academy	Education	54000
117	17	Sofia Fashion Hub	Fashion	81000
118	18	Ethan Review Central	Reviews	72000
119	19	Zara Cooking Magic	Cooking	128000
120	20	Noah Streaming Arena	Streaming	165000

1.3 Videos (5 rows)

Five videos across different channels with view/like counts and durations.

video_id	channel_id	title	upload_date	views	likes	duration (min)
1001	101	Best Laptop 2025	2025-01-10	500,000	45,000	15
1002	102	Learn SQL Basics	2025-02-12	750,000	62,000	25
1003	103	Top 10 Games	2025-03-05	1,200,000	98,000	18
1004	104	Trip to Japan	2025-03-15	650,000	54,000	30
1005	105	Relaxing Piano Music	2025-04-01	900,000	72,000	60

1.4 Comments (15 rows)

User comments across all 5 videos from January to April 2025.

comment_id	video_id	user_id	comment_text	comment_date
1	1001	1	Great tutorial!	2025-01-11
2	1003	2	Loved this video	2025-03-06
3	1004	3	Awesome gameplay	2025-03-16

comment_id	video_id	user_id	comment_text	comment_date
4	1002	4	Very helpful explanation!	2025-02-13
5	1005	5	Relaxing and peaceful music.	2025-04-02
6	1001	2	Thanks for the laptop review!	2025-01-12
7	1003	1	Top games list was amazing!	2025-03-07
8	1004	5	Japan looks beautiful!	2025-03-17
9	1002	3	Easy to understand SQL concepts.	2025-02-14
10	1005	4	Perfect music for studying.	2025-04-03
11	1001	5	This helped me choose a laptop.	2025-01-13
12	1003	4	Waiting for part 2!	2025-03-08
13	1004	2	Great travel vlog!	2025-03-18
14	1002	1	Excellent tutorial.	2025-02-15
15	1005	3	Loved the piano performance.	2025-04-04

1.5 Likes (15 rows)

Normalized like records — one row per user-video like event.

like_id	video_id	user_id	like_date
1	1001	1	2025-01-11
2	1001	2	2025-01-12
3	1002	3	2025-02-13
4	1002	4	2025-02-14
5	1003	5	2025-03-06
6	1003	1	2025-03-07
7	1004	2	2025-03-16
8	1004	3	2025-03-17
9	1005	4	2025-04-02
10	1005	5	2025-04-03

like_id	video_id	user_id	like_date
11	1001	3	2025-01-13
12	1002	5	2025-02-15
13	1003	2	2025-03-08
14	1004	1	2025-03-18
15	1005	2	2025-04-04

1.6 Subscriptions (20 rows)

User subscriptions to channels from January to March 2025.

subscription_id	user_id	channel_id	subscribe_date
1	1	101	2025-01-05
2	2	102	2025-01-10
3	3	103	2025-01-15
4	4	104	2025-01-20
5	5	105	2025-01-25
6	1	102	2025-02-01
7	2	103	2025-02-03
8	3	104	2025-02-05
9	4	105	2025-02-07
10	5	101	2025-02-10
11	1	103	2025-02-12
12	2	104	2025-02-15
13	3	105	2025-02-18
14	4	101	2025-02-20
15	5	102	2025-02-22
16	1	104	2025-03-01
17	2	105	2025-03-03
18	3	101	2025-03-05
19	4	102	2025-03-07
20	5	103	2025-03-10

1.7 Watch History (20 rows)

Records of viewing sessions — user, video, time watched, and date.

watch_id	user_id	video_id	watch_time (min)	watch_date
1	1	1001	12	2025-01-11
2	2	1002	20	2025-02-13
3	3	1003	15	2025-03-06
4	4	1004	25	2025-03-16
5	5	1005	45	2025-04-02
6	1	1002	18	2025-02-14
7	2	1003	10	2025-03-07
8	3	1004	22	2025-03-17
9	4	1005	50	2025-04-03
10	5	1001	14	2025-01-12
11	1	1003	16	2025-03-08
12	2	1004	28	2025-03-18
13	3	1005	55	2025-04-04
14	4	1001	13	2025-01-13
15	5	1002	21	2025-02-15
16	1	1004	24	2025-03-19
17	2	1005	48	2025-04-05
18	3	1001	11	2025-01-14
19	4	1002	19	2025-02-16
20	5	1003	17	2025-03-09

2. SQL Queries & Results

Query 1 — Top 3 Most-Viewed Videos

Retrieves the three videos with the highest view counts, ordered descending.

```
SELECT
  video_id,
  title,
  views
```

```
FROM videos
ORDER BY views DESC
LIMIT 3;
```

Result

video_id	title	views
1003	Top 10 Games	1,200,000
1005	Relaxing Piano Music	900,000
1002	Learn SQL Basics	750,000

Top 10 Games leads with 1.2M views — nearly 33% more than the second-ranked Relaxing Piano Music.

Query 2 — Top 3 Most-Liked Videos (from likes table)

Uses the normalized likes table with COUNT + JOIN to rank videos by actual like events recorded.

```
SELECT
  v.video_id,
  v.title,
  COUNT(l.like_id) AS total_likes
FROM videos v
JOIN likes l ON v.video_id = l.video_id
GROUP BY v.video_id, v.title
ORDER BY total_likes DESC
LIMIT 3;
```

Result

video_id	title	total_likes
1003	Top 10 Games	3
1001	Best Laptop 2025	3
1002	Learn SQL Basics	3

Note: With only 15 like records across 5 videos (3 each), the top 3 are tied. This demonstrates a difference from the denormalized likes column in the videos table.

Query 3 — All Videos by Like Count (LEFT JOIN)

Uses a LEFT JOIN to ensure all videos appear in results, including any with zero likes.

```
SELECT
  v.video_id,
  v.title,
  COUNT(l.like_id) AS total_likes
FROM videos v
LEFT JOIN likes l ON v.video_id = l.video_id
GROUP BY v.video_id, v.title
ORDER BY total_likes DESC;
```

Result

video_id	title	total_likes
1001	Best Laptop 2025	3
1002	Learn SQL Basics	3
1003	Top 10 Games	3
1004	Trip to Japan	3
1005	Relaxing Piano Music	3

All 5 videos have exactly 3 likes each from the sample data (15 total likes evenly distributed).

Query 4 — Total Views per Channel

Aggregates view counts at the channel level by joining channels to videos.

```
SELECT
  c.channel_id,
  c.channel_name,
  SUM(v.views) AS total_views
FROM channels c
JOIN videos v ON c.channel_id = v.channel_id
GROUP BY c.channel_id, c.channel_name
ORDER BY total_views DESC;
```

Result

channel_id	channel_name	total_views
103	Alex Gaming Zone	1,200,000
105	Mike Music Studio	900,000
102	Priya Coding Hub	750,000
104	Sarah Travel Vlogs	650,000
101	John Tech Reviews	500,000

Only 5 channels have uploaded videos so far. Alex Gaming Zone tops the chart with 1.2M total views. Channels without videos do not appear (INNER JOIN).

Query 5 — Most Active Watchers

Counts watch history sessions per user to identify the most engaged viewers.

```
SELECT
  u.user_id,
  u.username,
  COUNT(w.watch_id) AS total_watches
FROM users u
JOIN watch_history w ON u.user_id = w.user_id
GROUP BY u.user_id, u.username
ORDER BY total_watches DESC;
```

Result

user_id	username	total_watches
1	john_doe	4
2	priya_tech	4
3	alex_gaming	4
4	sarah_vlogs	4
5	mike_music	4

All 5 users with watch history have exactly 4 watch sessions each (20 total records, evenly distributed). The remaining 15 users have no watch history entries.

3. Key Insights

3.1 Content Performance

Metric	Value	Video / Channel
Highest views	1,200,000	Top 10 Games (Alex Gaming Zone)
Highest likes (raw)	98,000	Top 10 Games
Best like rate	9.0%	Best Laptop 2025 (45K likes / 500K views)
Longest video	60 min	Relaxing Piano Music
Shortest video	15 min	Best Laptop 2025
Avg video duration	29.6 min	Across 5 videos

3.2 Channel Rankings — Subscribers

Rank	channel_name	category	subscribers
1	Leo Gamer Pro	Gaming	245,000
2	Alex Gaming Zone	Gaming	210,000
3	Noah Streaming Arena	Streaming	165,000
4	Sam Music Beats	Music	156,000
5	Sarah Travel Vlogs	Travel	150,000
6	Lucas Live Streams	Entertainment	143,000
7	Oliver Tech Talks	Technology	134,000
8	Zara Cooking Magic	Cooking	128,000
9	John Tech Reviews	Technology	120,000
10	Kavya Daily Vlogs	Lifestyle	112,000

3.3 Engagement Overview

Table	Total Records	Notes
users	20	11 countries represented
channels	20	One per user; 15 categories
videos	5	Only 5 channels have uploaded
comments	15	3 comments per video on average
likes	15	3 likes per video (normalized)
subscriptions	20	Users 1–5 each have 4 subscriptions
watch_history	20	Users 1–5 each have 4 sessions

3.4 Notes & Observations

- The likes column in the videos table is denormalized. Always use the likes table for accurate analytics.
- Only users 1–5 appear in watch_history, comments, likes, and subscriptions — the remaining 15 users are inactive in the current dataset.
- Only 5 of 20 channels have videos. Channel-to-video JOIN queries will only return results for those 5 channels.

- Gaming is the top category by both subscribers (Leo Gamer Pro: 245K) and total views (Alex Gaming Zone: 1.2M views).
- Extend the dataset with more video uploads and user activity to produce richer analytics results.